amazon

Products Insights Using SQL

Data Analysis | Insights | Recommendations



- Completed Project
- GitHub Repository
- LinkedIn Post Done

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Why This Project?

From raw data to meaningful insights that support better business decisions.

WHY AMAZON

Amazon is a large e-commerce platform that generates a huge amount of sales and customer data.

WHAT I WANTED TO UNDERSTAND

I wanted to understand how SQL can be used to analyze this data, identify business problems, and discover opportunities for improvement.

WHAT I AIMED TO ACHIEVE

To improve my practical SQL skills while converting raw data into meaningful insights that can support better business decisions.

Skills I set out to build

Data cleaning

TRIM, NULL handling, type conversion

Joins

INNER and LEFT JOIN across 9 tables

Aggregations

SUM, COUNT, AVG, GROUP BY, HAVING

Business analysis

KPIs, CTEs and window functions

THE GOAL Raw data → Meaningful insights → Better business decisions

Project at a glance

TOTAL COMBINED REVENUE

$12,642,071

Revenue from all 21,629 orders, across every order status



Completed	$10.38M	82.1%	17,802 orders
Returned	$1.67M	13.2%	2,840 orders
Cancelled	$0.31M	2.4%	488 orders
In progress	$0.28M	2.2%	499 orders

$10.38M
completed revenue

17,802
completed orders

898
customers

765
products

Orders covered
2020-01-01 to 2024-07-30

What data do we have? 9 interconnected tables

customers

898

Customer master

MASTER

products

765

Product master: price, cogs

MASTER

sellers

54

Seller master

MASTER

category

6

Product categories

REFERENCE

orders

21,629

Order header: dates, status

TRANSACTION

order_items

21,629

Product line per order

TRANSACTION

inventory

765

Current stock per product

OPERATIONAL

payments

21,629

Payment status per order

OPERATIONAL

shipping

21,141

Delivery and return info

OPERATIONAL

KPI definitions used throughout

Revenue

SUM(quantity × price_per_unit)

AOV

Completed revenue ÷ completed orders

Historical CLTV

Cumulative completed revenue per customer

Gross profit

(price_per_unit − cogs) × quantity

Return rate

Returned units ÷ total units × 100

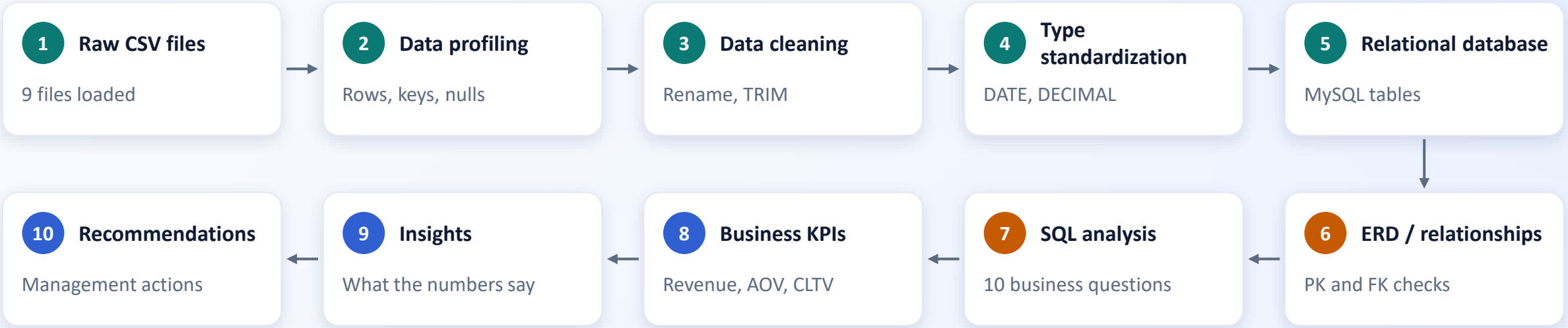
Stock risk

Products with stock < 10 units

Revenue, AOV, CLTV and profit use completed orders unless stated.

Row counts are from the supplied CSV files. Each order has 1 order_item row and 1 payment row; shipping has 488 fewer rows (cancelled orders).

Methodology: from raw CSV to recommendations



SQL techniques demonstrated

Basic SQL

- SELECT
- WHERE
- ORDER BY
- LIMIT

Aggregation

- SUM()
- COUNT()
- AVG()
- GROUP BY
- HAVING

Joins

- INNER JOIN
- LEFT JOIN

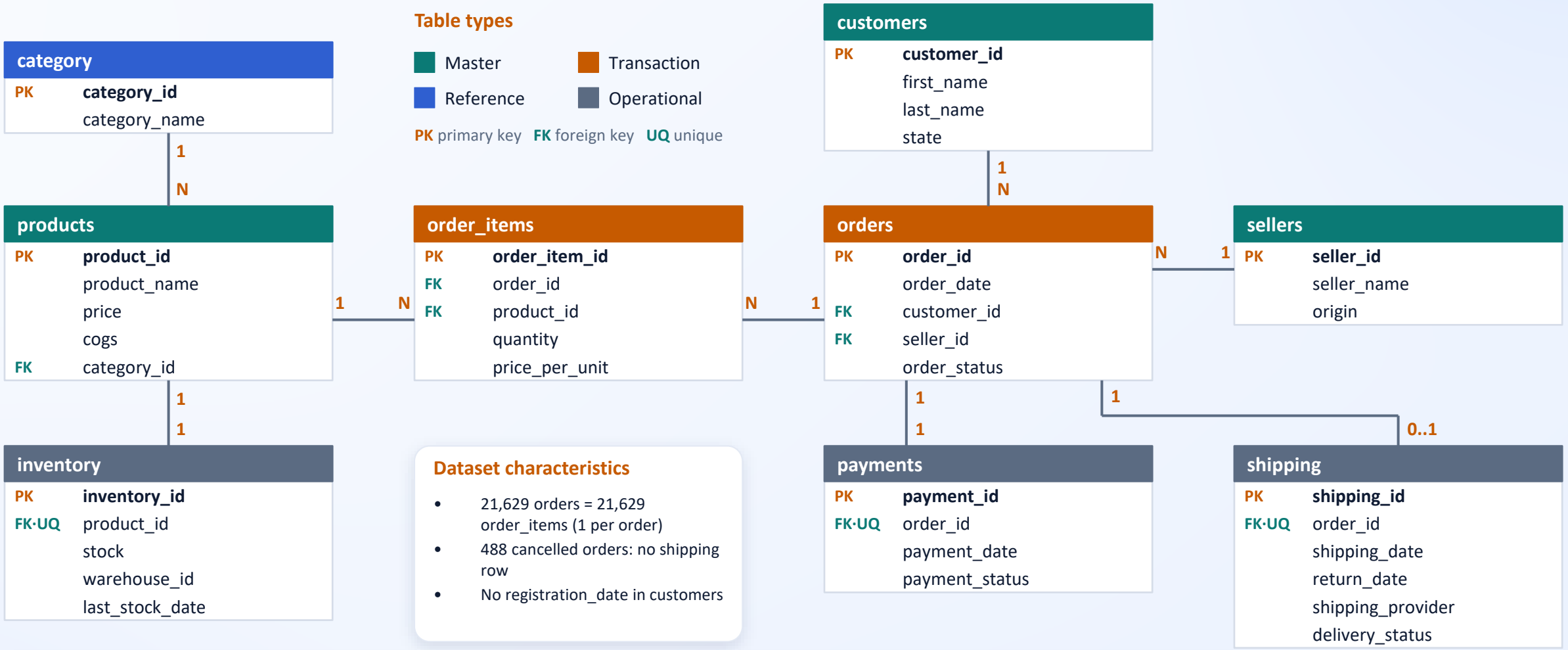
Advanced SQL

- CTEs
- CASE
- Window: LAG()
- Window: RANK()

Data cleaning

- TRIM()
- NULL handling
- Datatype conversion
- Date functions

Database Architecture & Entity Relationship Diagram



Dataset characteristics

- 21,629 orders = 21,629 order_items (1 per order)
- 488 cancelled orders: no shipping row
- No registration_date in customers

Key relationship observations

- Customers → Orders: 1:N
- Sellers → Orders: 1:N
- Categories → Products: 1:N
- Products → Order Items: 1:N
- Products → Inventory: 1:1 in supplied data
- Orders → Payments: 1:1 in supplied data
- Orders → Shipping: 0..1 in supplied data
- Orders → Order Items: 1:N by design; 1 row per order here

Is the data reliable? Keys are clean, types are not

Structure
9
tables analyzed

Customers
898
customer master rows

Products
765
product master rows

Orders
21,629
order header rows

Foreign-key validation
0 orphans
all 8 foreign keys resolve

Never-ordered customers
212
customers with no order

Never-sold products
15
never in order_items

Low-stock products
51
current stock < 10 units

Missing shipping records
488
orders with no shipping row

Missing registration_date
Yes
not in the customers table

Datatype mistakes and corrections

Source field	Found in CSV	MySQL type	Why
order_date	text	DATE	Date maths
payment_date	text	DATE	Date analysis
shipping_date	text	DATE	Delivery analysis
return_date	text, blank	DATE NULL	Return analysis
last_stock_date	text	DATE	Stock analysis
price	float	DECIMAL(10,2)	Money precision
cogs	float	DECIMAL(10,2)	Money precision
price_per_unit	float	DECIMAL(10,2)	Money precision
quantity	integer	SMALLINT	Unit counts
status fields	text + spaces	VARCHAR + TRIM	Clean labels

Cleaning + FK validation

```
UPDATE payments SET payment_status =
       TRIM(payment_status);
-- orders without a customer (expect 0)
SELECT COUNT(*) FROM orders o
LEFT JOIN customers c ON c.customer_id = o.customer_id
WHERE c.customer_id IS NULL;
```

Data mistakes found in the CSVs and how they were fixed

#	Mistake	Where	Evidence	Fix or handling
1	**Dates stored as text**	5 date columns in 4 tables	All load as text (YYYY-MM-DD), not DATE	Convert with STR_TO_DATE() to DATE
2	**Trailing spaces in labels**	payments.payment_status, shipping.delivery_status	2,840 rows each ("Refunded ", "Returned ")	TRIM() before grouping
3	**Column names with spaces**	customers."Customer ID", shipping."shipping providers"	Need quotes and break joins	Rename to customer_id, shipping_provider
4	**Spelling errors in labels**	payment_status, order_status	"Payment Successed" (18,301 rows), "Inprogress" (499)	Source text kept in queries; fix to Succeeded, In Progress
5	**Inconsistent text format**	category_name, shipping_provider, product_name	electronics vs Pet Supplies; dhl, fedex lowercase; 11 names with curly apostrophes	Standardise case; use utf8mb4
6	**Duplicate product names**	products.product_name	5 names used by 2 product_ids with different prices (Board Game 34.99 and 24.99)	Group by product_id, never by name
7	**Blank instead of NULL**	shipping.return_date	18,301 blank values (orders not returned)	Store as NULL (DATE NULL)
8	**Flat order lines**	order_items.order_item_id	Equals order_id in every row: 1 item per order	Keep the 1:N design; do not claim multi-item baskets
9	**Payment date has no lag**	payments.payment_date	Equals order_date in all 21,629 rows	Do not use it for payment-delay analysis
10	**Missing fields**	customers, order_items	No registration_date; price_per_unit always equals products.price	No cohort or price-change analysis possible
11	**Implausible margins**	products.cogs	Margin 55% to 95%, mean 74.7% (Board Game: cogs 3.15 on price 34.99)	Treat gross profit as indicative only
12	**Possible duplicate customers**	customers first_name, last_name	358 repeated full names (235 also share a state)	Cannot confirm; customer_id is the only key

Checks that passed 0 orphan keys | 0 invalid dates | 0 non-positive prices or quantities | cogs always below price | shipping 1 to 5 days after order

The 10 business questions this project answers

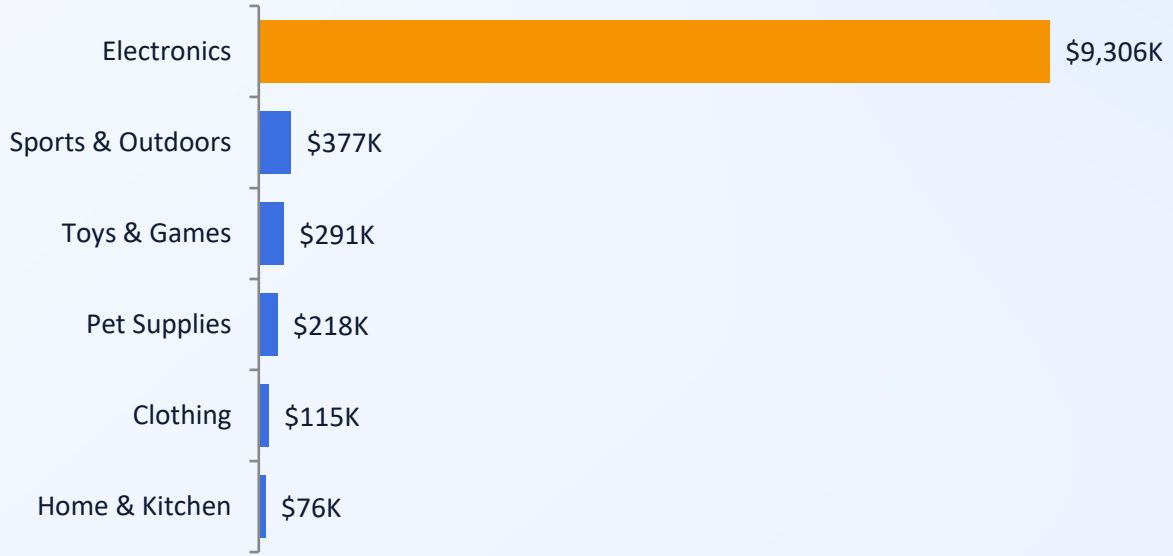
Q	Business question	Answer from the data	Slide
Q1	**Which product categories generate the most revenue?**	Electronics: 89.63% of completed revenue ($9.31M)	9
Q2	**Which products are the best sellers by revenue?**	Apple iMac Pro: $505K from 101 units	10
Q3	**Which states bring in the most revenue?**	Ohio and Texas: 90.2% of completed revenue	12
Q4	**What share of payments succeed, are refunded or fail?**	84.61% succeeded, 13.13% refunded, 2.26% failed	13
Q5	**Which products have a return rate above 20%?**	12 products with 50+ units; worst is DJI Mavic Air 2 (30.5%)	13
Q6	**How many customers have never placed an order?**	212 customers (23.6% of 898)	14
Q7	**Which products are running low on stock?**	51 products with stock below 10 units	15
Q8	**Which products have never been sold?**	15 products never appear in order_items	15
Q9	**How has revenue changed year over year?**	+34.7% in 2021, -5.9% in 2022, +4.5% in 2023; 2024 is partial	11
Q10	**Who are the most valuable customers (historical CLTV)?**	Yvonne Reed (Ohio): $89,029; average CLTV $19,967	14

What drives revenue? Electronics = 89.63%

Business question Q1: Which product categories generate the most revenue?

Q1 | Revenue by category, completed orders

```sql
WITH sales AS (
  SELECT p.category_id,
         oi.quantity * oi.price_per_unit AS rev
  FROM orders o
  JOIN order_items oi ON oi.order_id = o.order_id
  JOIN products p ON p.product_id = oi.product_id
  WHERE o.order_status = 'Completed'
)
SELECT c.category_name,
       SUM(s.rev) AS revenue,
       ROUND(100 * SUM(s.rev) /
         (SELECT SUM(rev) FROM sales), 2) AS pct
FROM sales s
JOIN category c ON c.category_id = s.category_id
GROUP BY c.category_name
ORDER BY revenue DESC;
```



89.63%

of completed revenue is Electronics

$10.38M

from 17,802 completed orders; AOV $583

$7.82M

gross profit, 75.3% margin

Category	Revenue	Share
Electronics	$9,306,136	89.63%
Sports & Outdoors	$377,191	3.63%
Toys & Games	$290,594	2.80%
Pet Supplies	$218,040	2.10%
Clothing	$114,607	1.10%
Home & Kitchen	$76,281	0.73%

Which products drive sales? Apple iMac Pro leads

Business question Q2: Which products are the best sellers by revenue?

Q2 | Top 10 products by completed revenue

```sql
SELECT p.product_id, p.product_name,
       SUM(oi.quantity) AS units,
       SUM(oi.quantity * oi.price_per_unit) AS revenue
FROM orders o
JOIN order_items oi ON oi.order_id = o.order_id
JOIN products p ON p.product_id = oi.product_id
WHERE o.order_status = 'Completed'
GROUP BY p.product_id, p.product_name
ORDER BY revenue DESC
LIMIT 10;
```

$505K

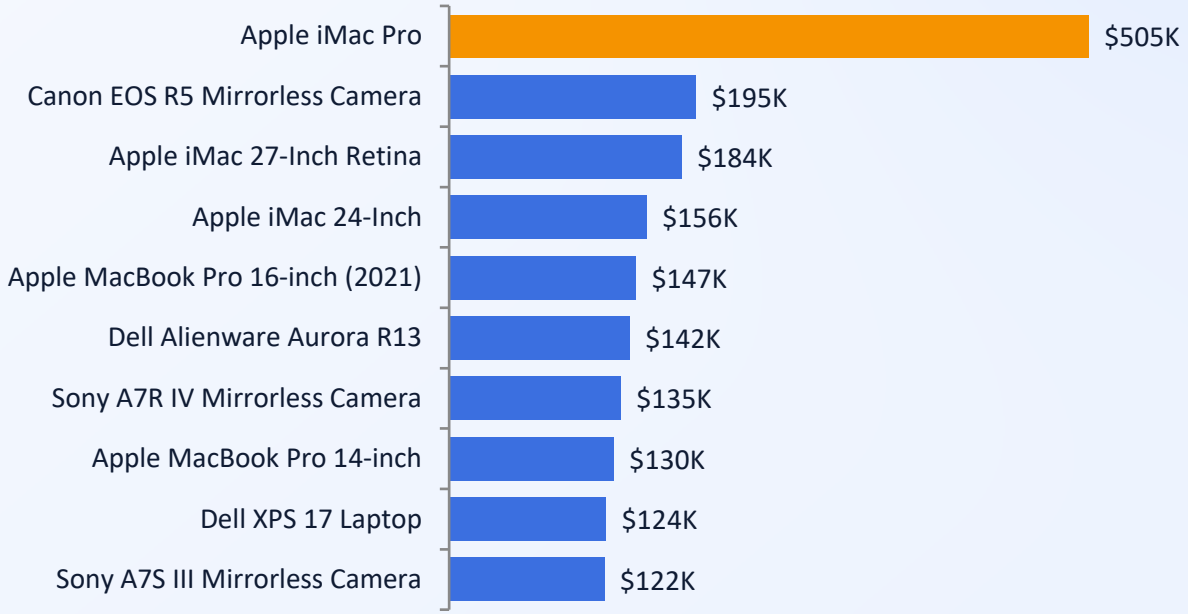
iMac Pro: 4.9% of all revenue

+$310K

lead over #2, Canon EOS R5

17.7%

of revenue from the top 10



Reading the result

- All top 10 products are Electronics
- iMac Pro sells 101 units at $4,999.99 each

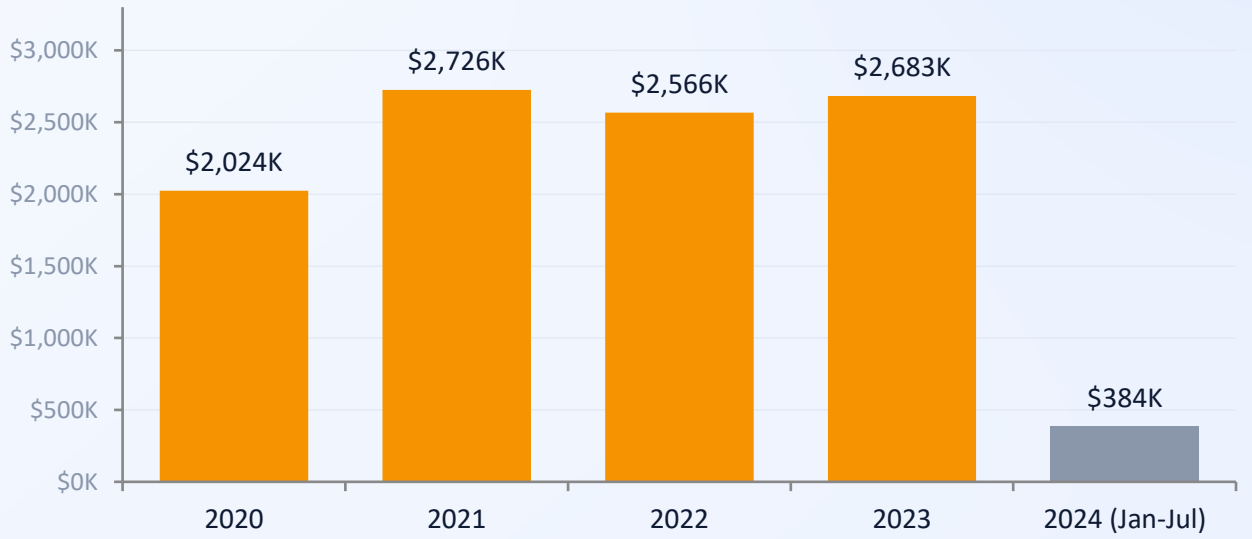
Top 5 products	Units	Revenue
Apple iMac Pro	101	$504,999
Canon EOS R5 Mirrorless Camera	50	$195,000
Apple iMac 27-Inch Retina	102	$183,599
Apple iMac 24-Inch	120	$155,999
Apple MacBook Pro 16-inch (2021)	59	$147,499

Revenue trend: steady to 2023, then a sharp monthly drop

Business question Q9: How has revenue changed year over year?

Q9 | Yearly revenue and growth (CTE + LAG)

```sql
WITH yearly AS (
  SELECT YEAR(o.order_date) AS yr,
         SUM(oi.quantity * oi.price_per_unit) AS revenue
  FROM orders o
  JOIN order_items oi ON oi.order_id = o.order_id
  WHERE o.order_status = 'Completed'
  GROUP BY YEAR(o.order_date))
SELECT yr, revenue,
       LAG(revenue) OVER (ORDER BY yr) AS prev_revenue,
       ROUND(100.0 * (revenue - LAG(revenue)
         OVER (ORDER BY yr)) / LAG(revenue)
         OVER (ORDER BY yr), 1) AS growth_pct
FROM yearly;
```



Check data capture before trusting 2024

- Data ends 2024-07-30: 2024 is a partial year, so the -85.7% is not a real decline.
- Orders per month (all statuses) fall from about 290 in Nov-Dec 2023 to 100-110 from Mar 2024.
- Completed revenue falls from a $224K monthly average in 2023 to under $25K a month, Mar-Jul 2024.

Year	Revenue	Prev year	Growth
2020	$2,023,678	-	-
2021	$2,725,931	$2,023,678	+34.7%
2022	$2,566,081	$2,725,931	-5.9%
2023	$2,682,791	$2,566,081	+4.5%
2024	$384,369	$2,682,791	-85.7%

Where is revenue concentrated? Ohio and Texas

Business question Q3: Which states bring in the most revenue?

Q3 | Top 5 states by completed revenue

```sql
SELECT cu.state,
       SUM(oi.quantity * oi.price_per_unit) AS revenue
FROM orders o
JOIN customers cu ON cu.customer_id = o.customer_id
JOIN order_items oi ON oi.order_id = o.order_id
WHERE o.order_status = 'Completed'
GROUP BY cu.state
ORDER BY revenue DESC
LIMIT 5;
```

90.2%

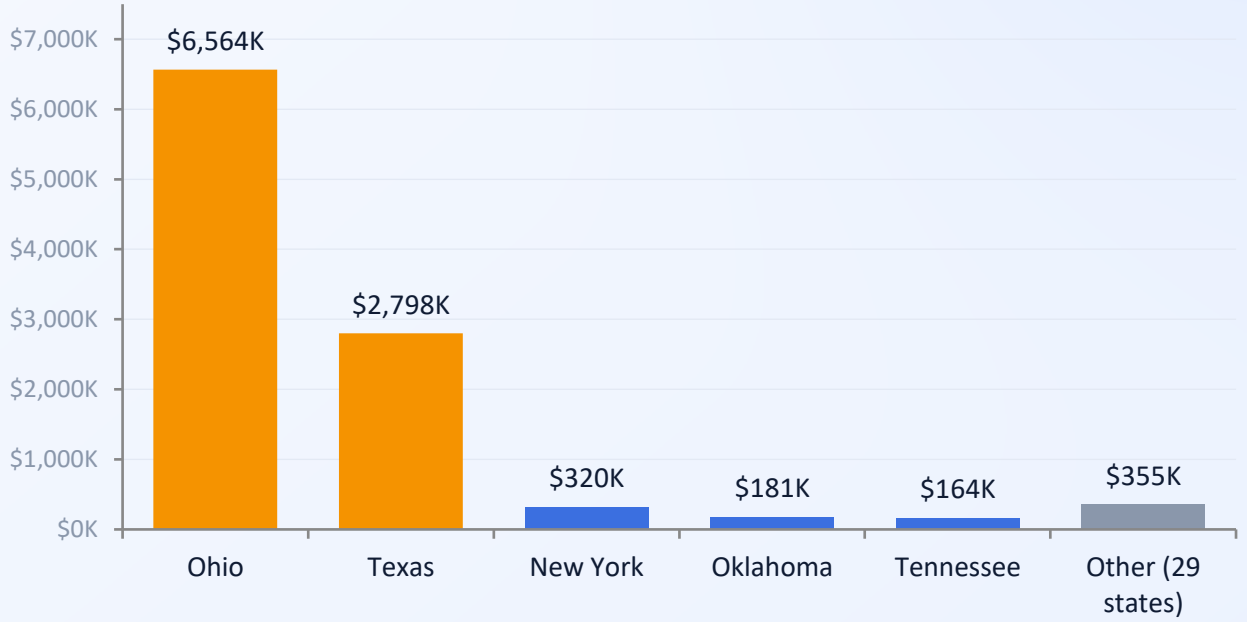
Ohio + Texas share

63.2%

Ohio: $6.56M

26.9%

Texas: $2.80M



State (top 5)	Revenue	Share
Ohio	$6,563,975	63.22%
Texas	$2,798,080	26.95%
New York	$320,286	3.08%
Oklahoma	$181,456	1.75%
Tennessee	$163,920	1.58%

Reading the result

- Only 34 of 49 customer states have completed sales
- Ohio and Texas together are 90.2% of revenue
- The other 29 states add only $355K combined

Payments and returns: 13.13% refunded

Business question Q4: What share of payments succeed, are refunded or fail? | Q5: Which products have a return rate above 20%?

Q4 | Payment status distribution

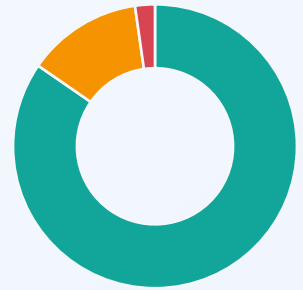
```sql
SELECT payment_status,
       COUNT(*) AS payments,
       ROUND(100.0 * COUNT(*) /
         (SELECT COUNT(*) FROM payments), 2) AS pct
FROM payments
GROUP BY payment_status
ORDER BY payments DESC;
```

Q5 | Products with return rate above 20% (50+ units)

```sql
WITH r AS (
  SELECT p.product_id, p.product_name,
         SUM(oi.quantity) AS units,
         SUM(CASE WHEN o.order_status = 'Returned'
             THEN oi.quantity ELSE 0 END) AS returned
  FROM orders o
  JOIN order_items oi ON oi.order_id = o.order_id
  JOIN products p ON p.product_id = oi.product_id
  GROUP BY p.product_id, p.product_name)
SELECT product_id, product_name, units, returned,
       ROUND(100.0 * returned / units, 2) AS return_pct
FROM r
WHERE units >= 50 AND 100.0 * returned / units > 20
ORDER BY return_pct DESC;
```

How it works: the CTE counts total and returned units per product. The outer query keeps products with 50+ units and a return rate above 20%.

Payment status



Refunded = Returned

2,840 refunded payments = 2,840 returned orders

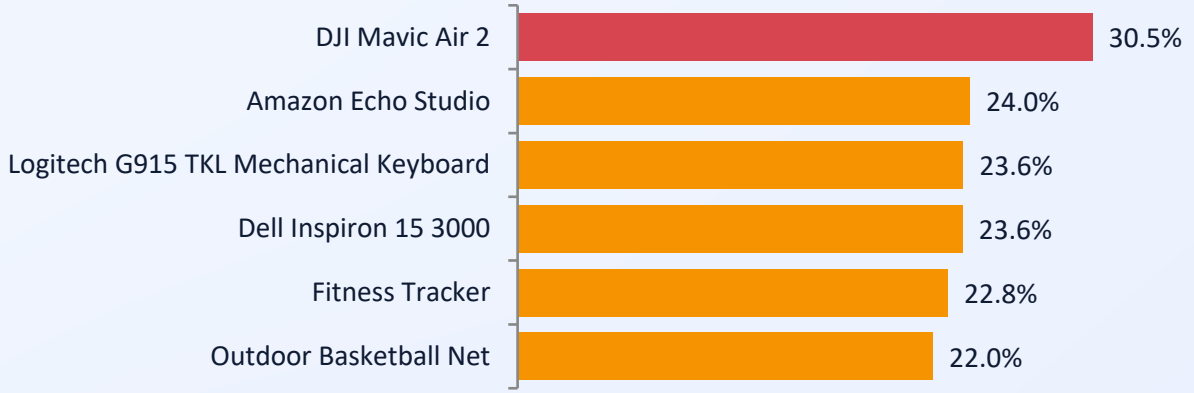
Failed = Cancelled

488 failed payments = 488 cancelled orders

Return rate 12.79%

of all units; 12 products (50+ units) exceed 20%

Highest return rates (50+ units sold)



Q6 Q10

Customers: 212 have never ordered

Business question Q6: How many customers have never placed an order? | Q10: Who are the most valuable customers (historical CLTV)?

Q6 | Customers with no order (LEFT JOIN + IS NULL)

```sql
SELECT COUNT(*) AS never_ordered_customers
FROM customers cu
LEFT JOIN orders o ON o.customer_id = cu.customer_id
WHERE o.order_id IS NULL;
```

Q10 | Top 5 customers by historical CLTV

```sql
WITH cltv AS (
  SELECT o.customer_id,
         SUM(oi.quantity * oi.price_per_unit) AS ltv
  FROM orders o
  JOIN order_items oi ON oi.order_id = o.order_id
  WHERE o.order_status = 'Completed'
  GROUP BY o.customer_id)
SELECT cu.first_name, cu.last_name, cu.state, c.ltv,
       RANK() OVER (ORDER BY c.ltv DESC) AS ltv_rank,
       (SELECT ROUND(AVG(ltv), 2) FROM cltv) AS avg_ltv
FROM cltv c
JOIN customers cu ON cu.customer_id = c.customer_id
ORDER BY ltv_rank
LIMIT 5;
```

Reading the results

- 212 of 898 customers (23.6%) never ordered
- 686 ordered; 520 have completed-order revenue

Top 5 customers by historical CLTV

Rank	Customer	State	CLTV
1	Yvonne Reed	Ohio	$89,029
2	Mia Reed	Ohio	$82,350
3	Fred Davis	Texas	$82,179
4	Quinn Davis	Ohio	$79,205
5	Nathan Lee	Texas	$77,137

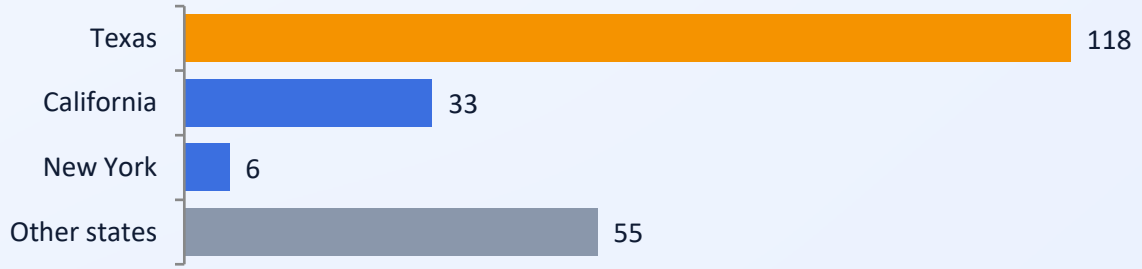
212

never-ordered customers; 118 in Texas, 33 in California

$19,967

average CLTV across 520 customers with completed orders

Where the 212 never-ordered customers live



Q7 Q8

Operations: low stock and unsold products

Business question Q7: Which products are running low on stock? | Q8: Which products have never been sold?

Q7 | Low-stock products (stock < 10)

```sql
SELECT p.product_name, i.stock
FROM inventory i
JOIN products p ON p.product_id = i.product_id
WHERE i.stock < 10
ORDER BY i.stock, p.product_name;
```

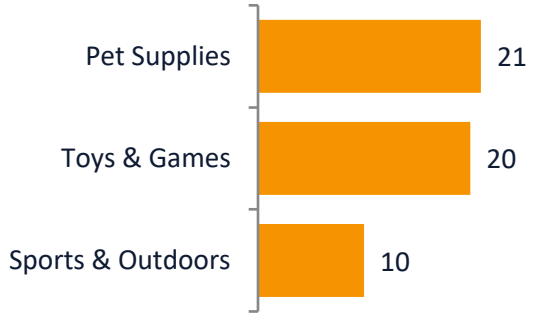
Q8 | Products that never appeared in order_items

```sql
SELECT p.product_id, p.product_name, p.price
FROM products p
LEFT JOIN order_items oi
       ON oi.product_id = p.product_id
WHERE oi.order_item_id IS NULL
ORDER BY p.price DESC;
```

Keep these separate

- 51 low-stock = current stock below 10 units (supply risk)
- 15 never-sold = no row in order_items (demand risk)
- 3 products appear in both lists
- All 51 low-stock products are in Pet Supplies, Toys & Games or Sports & Outdoors

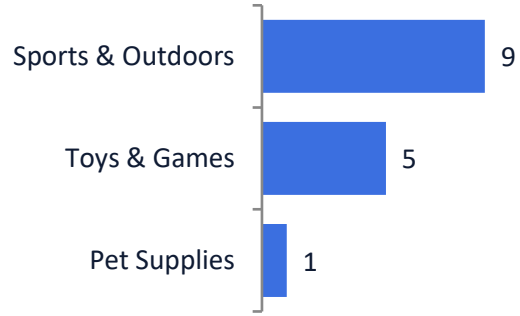
51 **Low stock**
stock < 10, by category



Lowest stock

- Archery Set: 1 unit
- Children's Puzzle: 1 unit
- Kid's Beach Tent: 1 unit
- Pet GPS Tracker: 1 unit
- Pet Water Fountain: 1 unit
- 7 products are at 1 unit

15 **Never sold**
no order_items row



Highest-priced unsold

- Golf Cart: $249.99
- Inflatable Obstacle Course: $149.99
- Lightweight Camping Tent: $79.99
- Baseball Batting Tee: $59.99
- Remote-Controlled Car Set: $54.99

Top 5 insights from the data

1	89.63%	**Electronics is the revenue engine** $9.31M of $10.38M completed revenue. No other category is above 4%.	Source: Q1
2	90.2%	**Two states carry the business** Ohio (63.2%) and Texas (26.9%) generate nearly all revenue. The other 29 states add only $355K.	Source: Q3
3	12.79%	**One in eight units comes back** 12 products with 50+ units have return rates above 20%. Refunded payments match returned orders exactly (2,840).	Source: Q4, Q5
4	212	**Almost a quarter of customers never buy** 23.6% of 898 customers have never placed an order, and 118 of them are in Texas.	Source: Q6
5	-65%	**2024 activity dropped sharply: real decline or missing data?** Data ends 30 Jul 2024. Orders fall from about 290 a month (Nov-Dec 2023) to about 100 (Mar-Jul 2024). Investigate whether this is a real decline or incomplete data.	Source: Q9

What should management do?

1

Plan inventory

51 products under 10 units

Set reorder points for Pet, Toys and Sports items, starting with the 7 products at 1 unit.

Based on Q7

2

Investigate returns

12 products above 20%

Among products with 50+ units sold. Review quality and listing accuracy, starting with DJI Mavic Air 2 at 30.5%.

Based on Q5

3

Activate customers

212 never ordered

Run a first-order campaign; 118 of them are in Texas.

Based on Q6, Q10

4

Review unsold products

15 never sold

Check price, visibility and listings, then relist or retire. 9 are Sports & Outdoors.

Based on Q8

5

Improve data capture

No registration_date

Add registration dates, fix text-typed dates and label typos, and verify whether the 2024 order drop is real or missing data.

Based on data audit

Concentration risk to track Electronics = 89.63% of completed revenue | Ohio + Texas = 90.2% | Top 10 products = 17.7%

CONCLUSION

Thank you for your time

Clean data, clear SQL and focused actions turn raw CSV files into business decisions.

Questions and feedback are welcome.

Divyesh Lodha | Data Analyst | Data with Divyesh

1

Revenue is concentrated

Electronics = 89.63% of revenue; Ohio and Texas = 90.2%.

2

Operations need attention

51 low-stock products, 15 never sold, 12 with returns above 20%.

3

Customers are untapped

212 customers (23.6%) have never placed an order.

4

Data quality needs fixing

12 data mistakes found, and the 2024 drop must be verified.